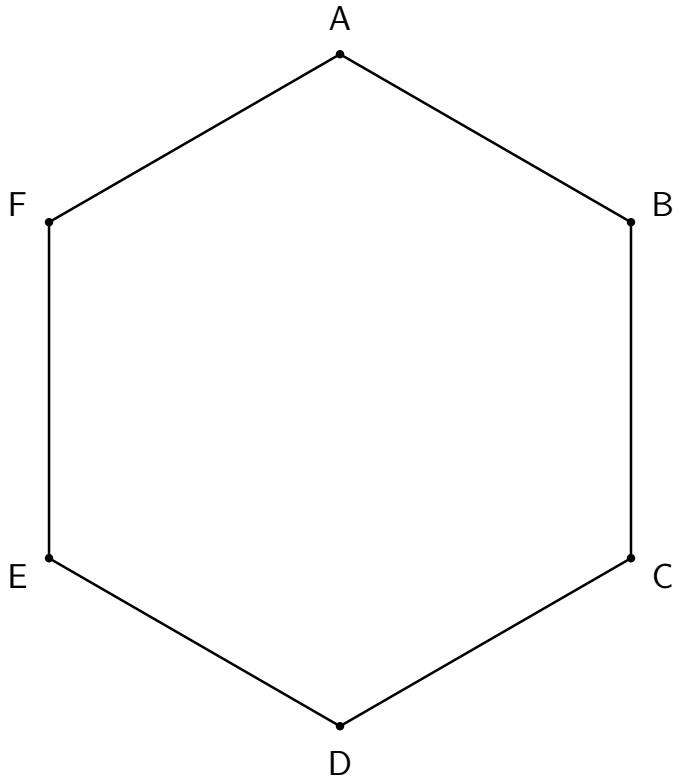
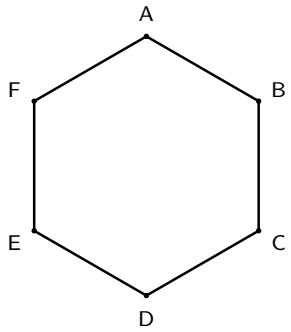
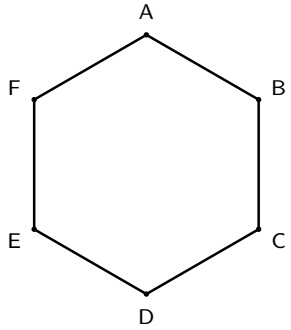
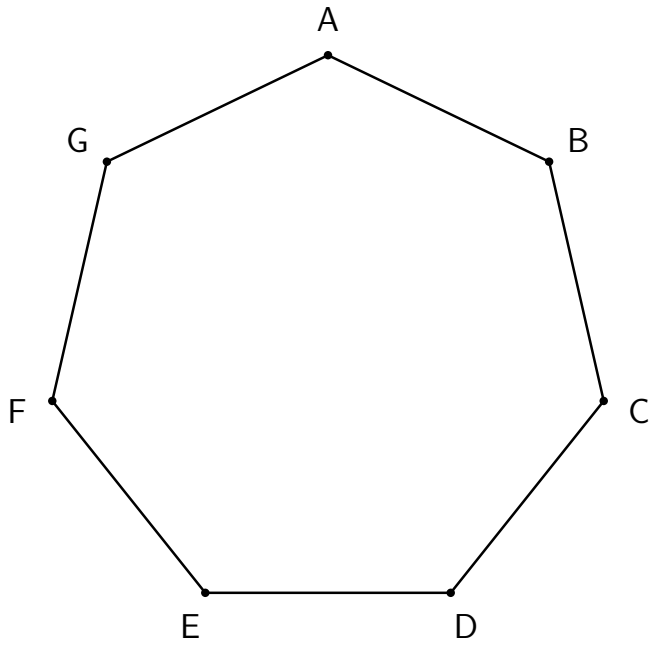
Fill the whole polygon with triangles, using straight lines between printed corners. Lines inside a polygon may meet at corners but may not cross. Keep the corner labels fixed, even if you turn the page. Use tracing paper to change printed lines.

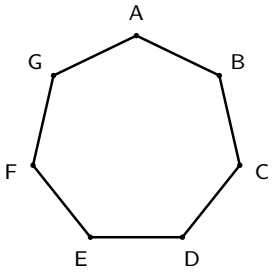
Problem 1: Fill each polygon in two different ways. Can the two fillings of one polygon have different numbers of triangles?

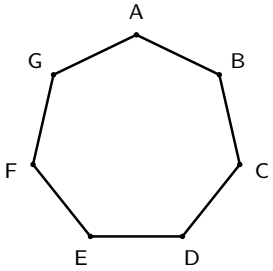




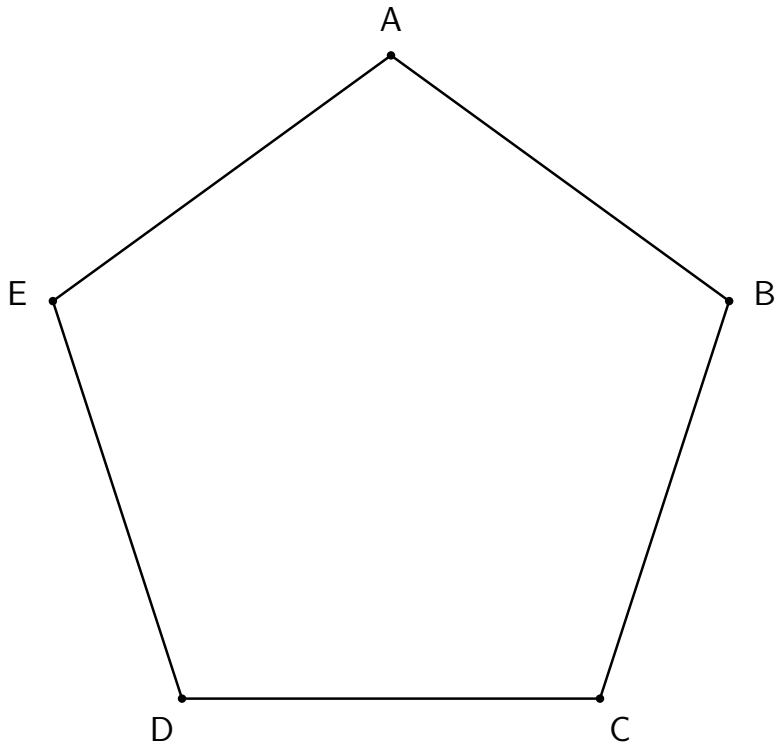


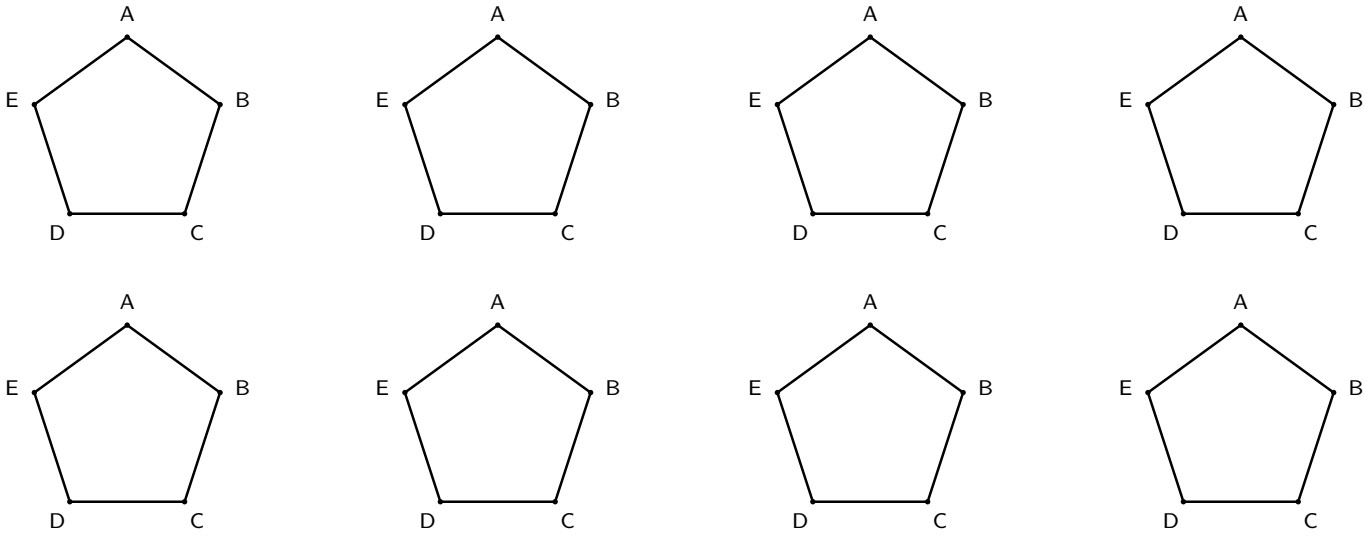




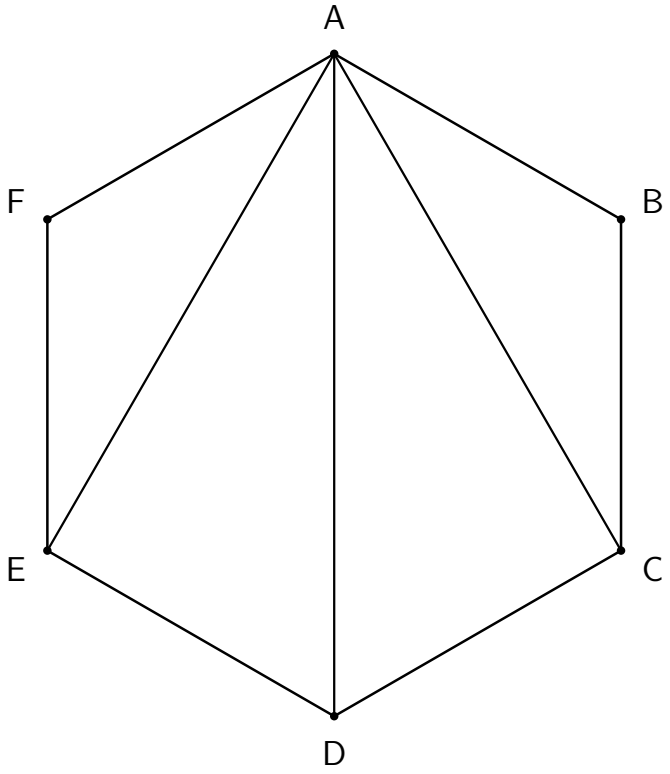


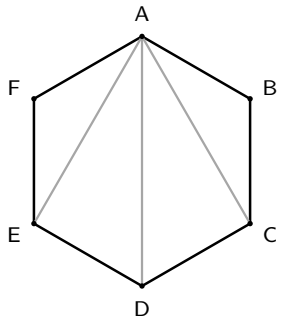
Problem 2: Find every filling of this five-corner polygon. How do you know none are missing?

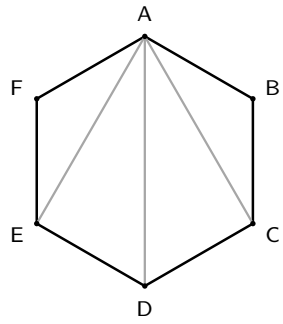


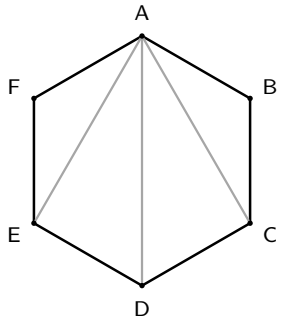


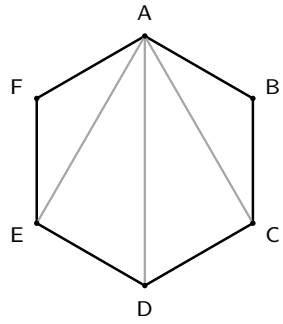
Problem 3: Erase one inside line and replace it with a different line so the polygon stays filled with triangles. Find every result of one such change from each drawing.

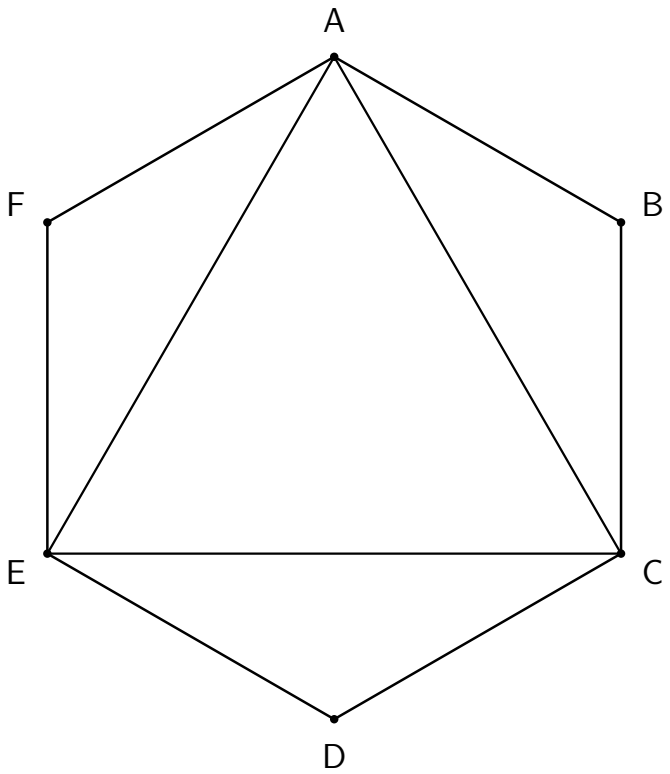


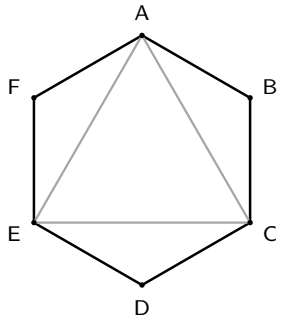


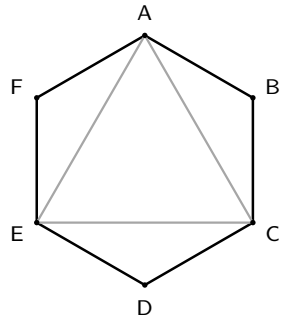


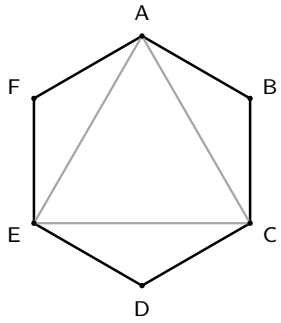


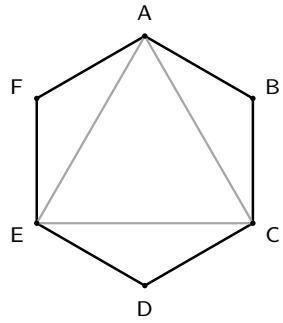




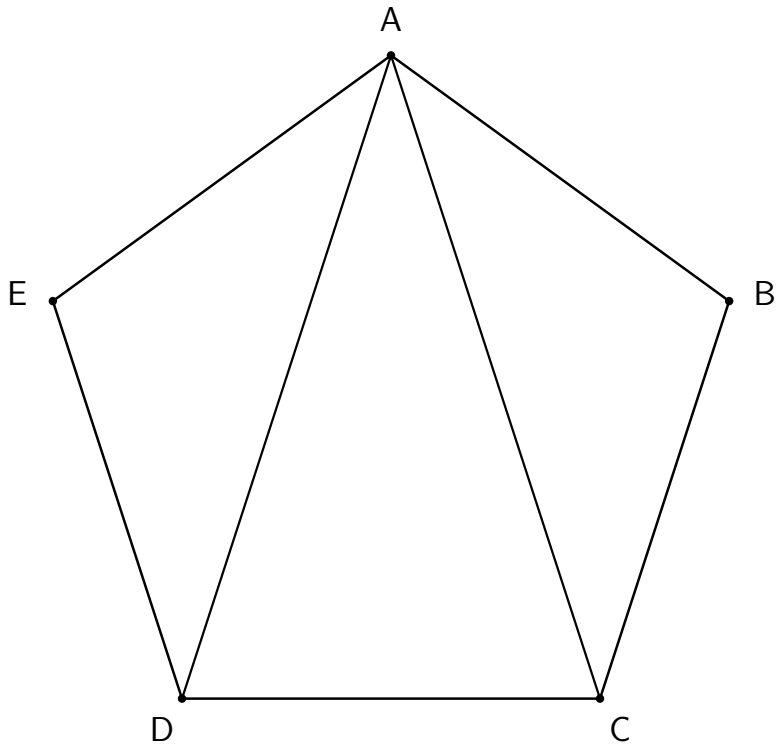


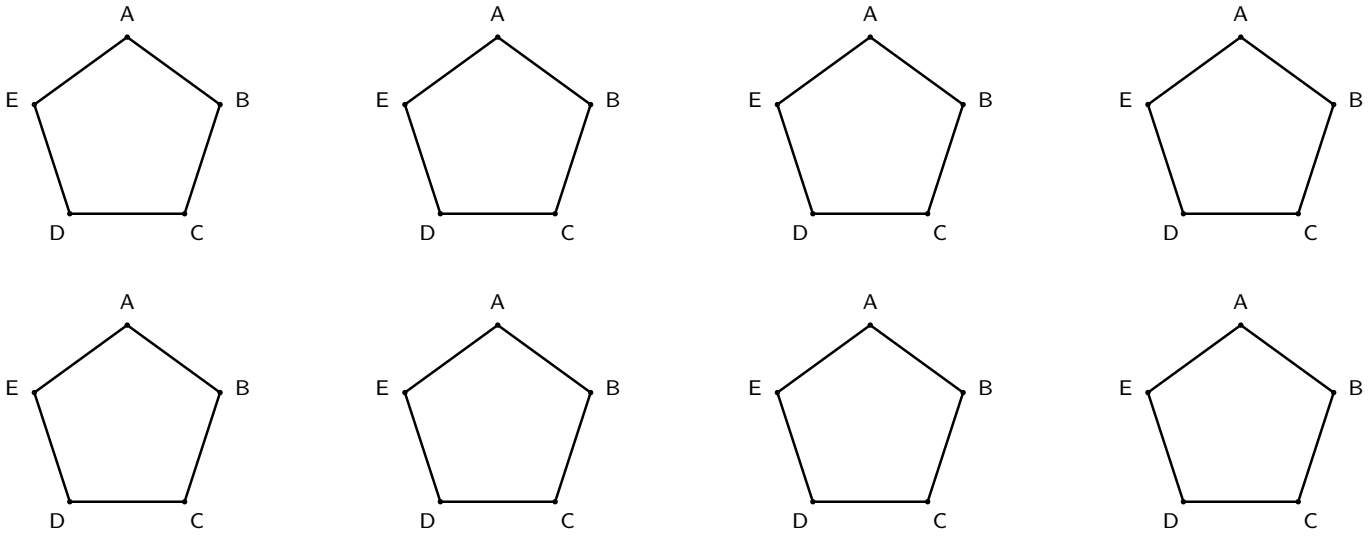




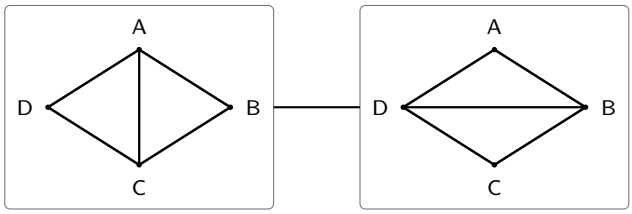


Problem 4: Start with this filling. Change one inside line at a time, keeping the polygon filled with triangles, and get back to this filling after an odd number of changes. Show every filling along your route.

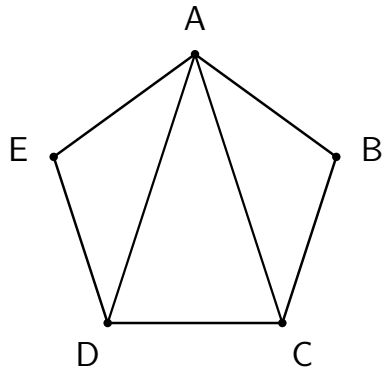


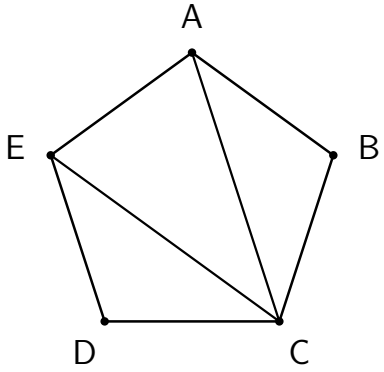


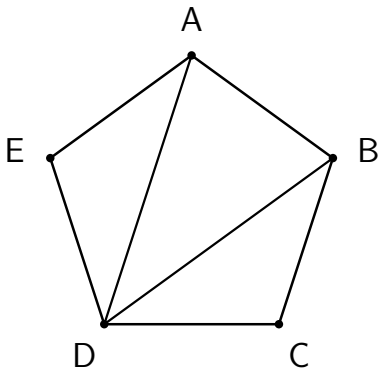
Problem 5: A change of one inside line that keeps the polygon filled with triangles is called a flip. Join two drawings when one flip changes one into the other. Include every possible join.

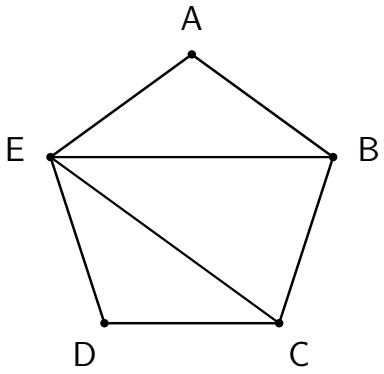


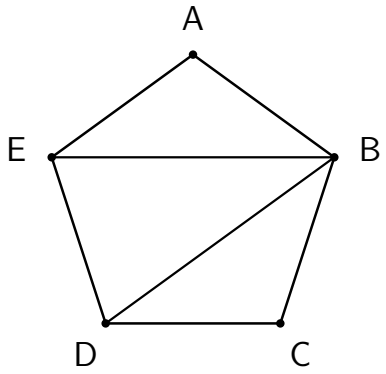
Example: erase AC, draw BD. The join connects whole fillings; it is not an inside line.



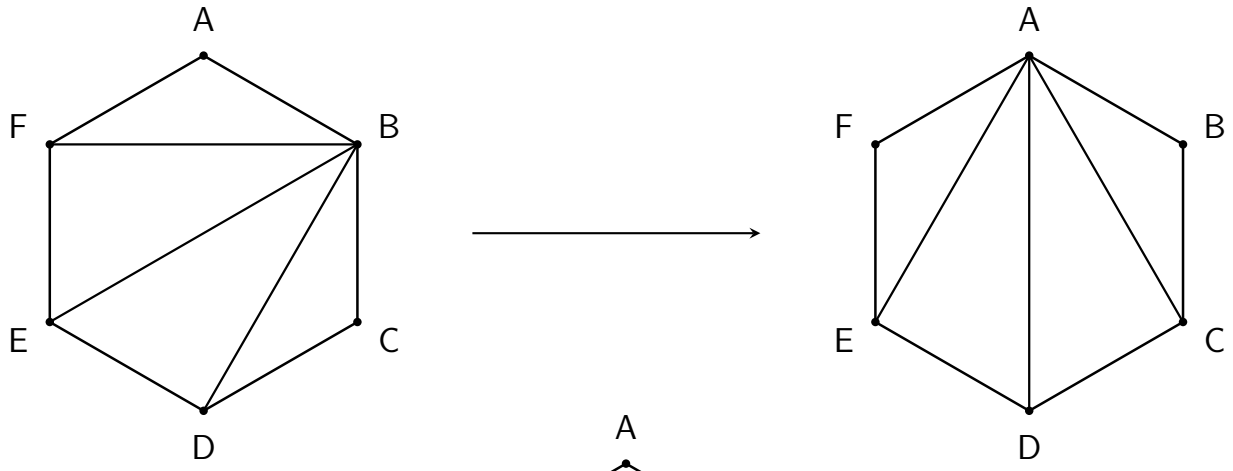


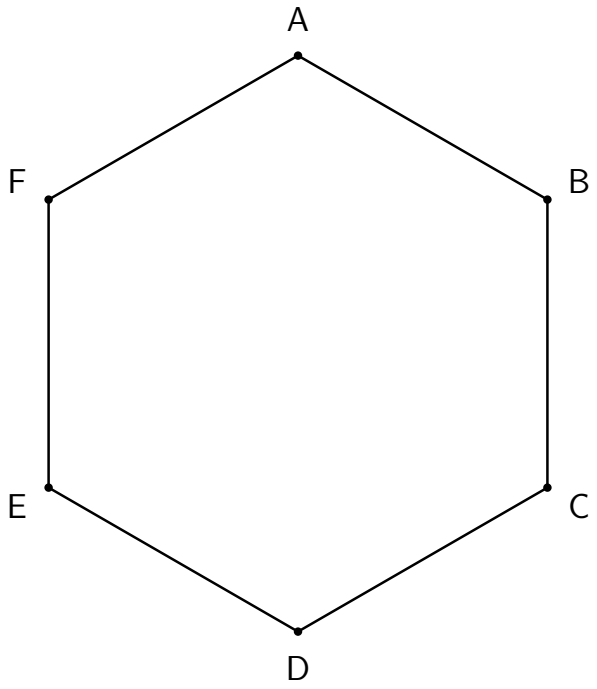






Problem 6: Change the first filling into the second using as few flips as you can. Find another route with a different number of flips, without visiting any filling twice.

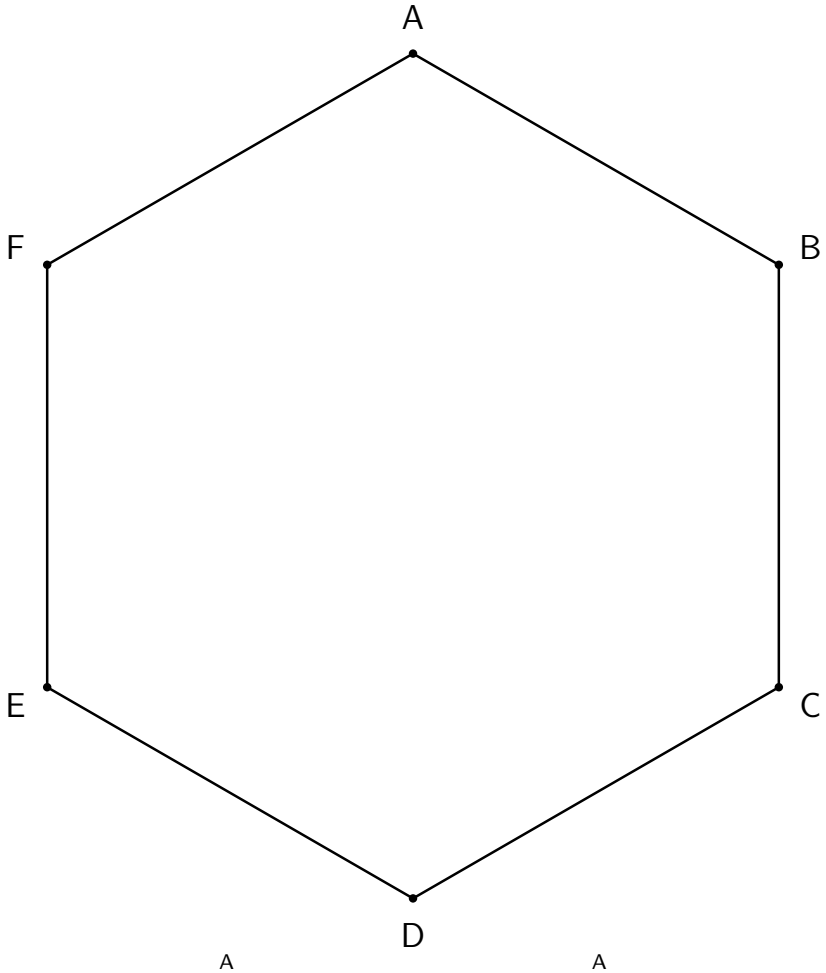


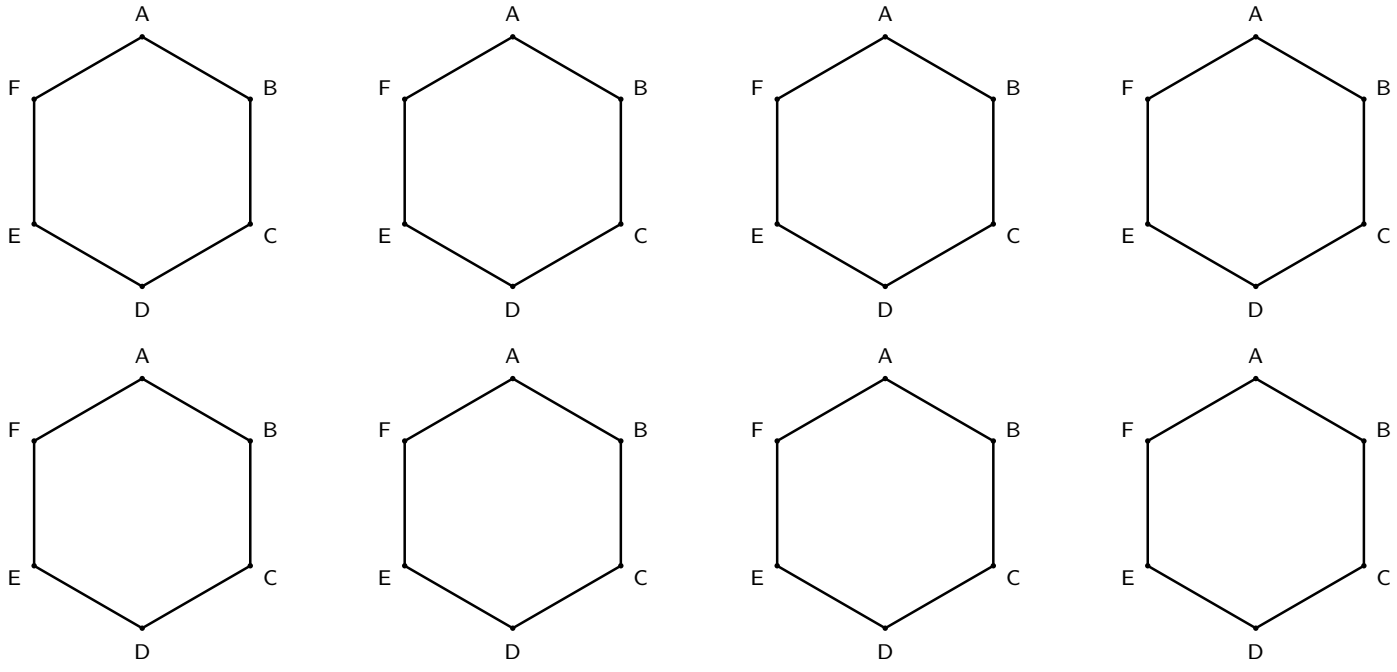


Fewest flips

Another route

Problem 7: Find every filling of this six-corner polygon. How do you know none are missing or repeated?





Problem 7 (continued)

